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The Editors
Nature Medicine

Dear Editors,

We are pleased to submit our manuscript, "Do We Need the Largest LLMs for Every Clinical Decision?", for consideration as an Article in *Nature Medicine*.

The best open-weight clinical language models are also among the largest, and running a 70-billion-parameter model at full precision requires multi-GPU infrastructure that many hospitals, and most health systems in lower-resource settings, do not have. Compression can shrink these models, but it is currently applied as a single global precision choice with no guarantee about which individual clinical decisions it changes. That gap is precisely what keeps compressed models out of settings where a silent change to a single decision is unacceptable. Our work addresses this deployment barrier directly.

We introduce CERTA, a decision-level statistical certificate that decides, for each input, whether a quantized model may stand in for its full-precision reference and escalates the decisions it cannot certify back to that reference. Rather than asking whether a compressed model is "good enough" on average, CERTA provides a finite-sample, per-decision guarantee on its disagreement with the trusted full-precision model, turning compression from a fixed engineering choice into an auditable, risk-controlled deployment policy.

Evaluating eleven general and domain-adapted models from three families (7–72B) on 2,500 clinical-verification decisions constructed from real de-identified patient contexts (2,484 patients), with patient-disjoint calibration and testing, we find that eight-bit quantization certified at least 99.8% of decisions for every model. Under a class-conditional guarantee, a four-bit scheme reduced the certified footprint to about a quarter of the full-precision size for two 72B models, moving them from a measured 142–146 GB across two 80 GB GPUs to 41–43 GB, within the capacity of a single 48 GB GPU. Independent disease named-entity benchmarking against human annotations confirmed that the deployed configurations largely preserved accuracy on the external task, and not merely agreement with the reference.

We believe this is a good fit for *Nature Medicine* because the contribution is clinical rather than purely methodological. It offers a concrete, statistically principled route to running capable clinical language models safely on hardware that ordinary care and research settings can afford, with an explicit and tunable bound on decision-level risk and a transparent escalation mechanism for the cases the small model cannot handle. This has direct implications for equitable access to clinical AI, for on-premises deployment where data cannot leave the institution, and for the governance of model updates in clinical workflows.

The study uses only already-published, de-identified case-report summaries (the public PMC-Patients corpus) and involved no interaction with human subjects, so it required no additional ethical approval or informed consent. All code and the recorded decision-level outputs needed to reproduce every result are

provided in a public repository. The manuscript is original, is not under consideration elsewhere, and both authors have approved the submission. We declare no competing interests.

Thank you for considering our work. We would be glad to provide any further information.

Sincerely,

Ali H. Lazem, on behalf of both authors